

Part A

1.

a. $R_T = R_1 + R_2 + R_3$

$$R_T = 20 + 30 + 50$$

$$= 100 \text{ ohms}$$

b. $1/R_T = 1/R_1 + 1/R_2 + 1/R_3$

$$1/R_T = 1/20 + 1/100 + 1/50$$

$$= 2/25$$

$$R_T = 25/2 = 12.5 \text{ ohms}$$

2.

a.

$$I = V/R$$

$$V = 125\text{V} \quad R = 100 \text{ ohms}$$

$$I = 125\text{V}/100 \text{ ohms} = 1.25\text{A}$$

b.

$$V = 125\text{V} \quad R = 12.5 \text{ ohms}$$

$$I = 125/12.5 = 10\text{A}$$

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a.

$$I_t = I_1 = I_2$$

$$I = 1.25\text{A}$$

Current through each resistor is 1.25A

b.

$$I_1 = 125/20 = 6.25\text{A}$$

$$I_2 = 125/100 = 1.25\text{a}$$

$$I_3 = 125/50 = 2.5\text{A}$$

4.

a.

$$I = V/R$$

$$V = I * R$$

$$V_1 = 1.25A * 20$$

$$V_1 = 25V$$

$$V_2 = 1.25A * 30$$

$$V_2 = 37.5V$$

$$V_3 = 1.25A * 50$$

$$V_3 = 62.5V$$

b.

$$V_T = V_1 = V_2 = V_3$$

$$V = 125V$$

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a.

$$\text{Power } P = V * I$$

$$P_1 = 25V * 1.25A = 31.25W$$

$$P_2 = 37.5V * 1.25A = 46.875W$$

$$P_3 = 62.5V * 1.25A = 78.125W$$

b.

$$P = V * I$$

$$P_1 = 125V * 6.25A = 781.25W$$

$$P_2 = 125V * 1.25A = 156.25W$$

$$P_3 = 125 * 2.5A = 312.5W$$